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Analysis of a Roll Cage Design against Various Impact Load and Longitudinal Torsion for Safety

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Abstract. This study is aimed at carrying out a finite element analysis on a roll cage of a vehicle used for student competitions around the world. The design process was carried out using a CAD package and analysis was done using finite element analysis software, ANSYS. Impact test was carried out on the roll cage under different conditions and the effect on the structural members are studied and discussed. The results are very much helpful and assure that the car is in safe condition in case of a crash or accident. Finite Element Analysis (FEA) is also carried out with the frame against torsional loading. Two different materials were considered for the analysis. The results of displacement of frame, nodal solution, element solution and bending moment are discussed.

Introduction

The need for design and analysis of the roll cage of a racing car used by students during competitions are to addressed because of the failure of structural members which leads to accidents. The structure of a roll cage must be light weight and rigid. Also, it must act as the central mounting bracket for all other systems and components. The design strength of the structure cannot be found directly from any empirical formulas instead has to be found out from the reviews of the drivers. Vehicle body is divided into two parts; roll cage and bodywork. The conventional roll cage looks similar to a truss and is welded at joints. This frame protects its occupants from an accident especially in case of a roll over. A chassis which can absorb high energy impacts whilst controlling the rate of deceleration will increase the likelihood of drivers surviving a crash without injury [1].

The design procedure begins with the research on past designs and developing new ideas. After the initial design has been chosen, for further development, it is subjected to finite element modelling and optimization. The next step is the engineering of the frame to a usable position. The structure is then tested to obtain some useful information after which the designer may change the decisions made during the design process. The most important point to be considered while designing the frame is that it must be reliable, economical and highly adjustable[2]. A three dimensional finite model was formulated with pipe16 elements and elastic supports to allow for simulation of various levels of load transfer [3].

The first step in designing a roll cage is the knowledge of various loads acting on it. The main modes for an automotive roll cage are given in [4] as: (i) Longitudinal torsion (ii) Vertical bending (iii) Lateral bending (iv) Horizontal lozenging. Torsion loads is caused due to the action of loads on the oppositely opposed corners of the car. The frame can be considered to be as torsion spring connecting the two ends at which the suspension loads act. Torsional loading affects both the handling as well as the performance of the car. Some experimental methods were presented with an emphasis on the whole-car torsion procedure [5]. A flexible chassis reduces the ability of the race engineer to translate a change in front to rear roll stiffness ratio into a change in front to rear load transfer distribution [6]. In reality the car will impact with deformable objects, such as tyre, barriers, cones or other cars [7]. Static analysis done in a study showed that deflections decreases with increase in stiffness when the back of the chassis was loaded [8]. Finite element design tool has significantly improved both the standards of engineering designs and the methodology of the design process in many industrial applications [9].

DESIGN PROCEDURE

Initially design were studied by making use of the previous data available. The branches in the roll cages are used to distribute loads from the engines into the unit as well as providing protection from rollover. 3D modelling of the roll cage was carried out using *Solid Works* software and is shown in Fig. 1. This model was then imported to finite element package ANSYS by converting the model into IGES format. IGES is a neutral data format that allows the transfer or exchange of data among the computer aided systems. It was then checked for any data losses which might have occurred during the importing. Appropriate element type was selected to represent the model. The various data regarding the material properties was entered. Then the model was meshed. as shown in Fig. 2.

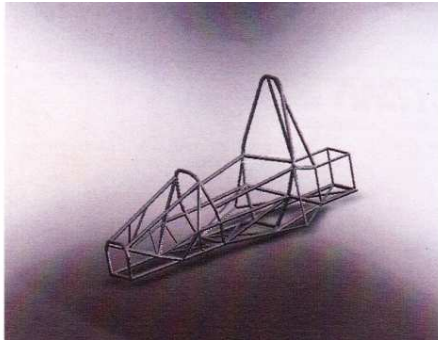


Fig. 1 CAD Model of the Roll Cage

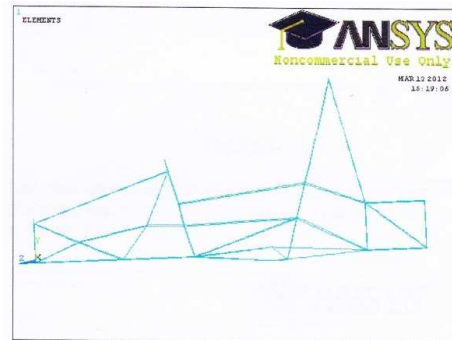


Fig. 2 Finite Element Model

Impact Test Analysis

Two types of analysis, namely front and side impact was carried out on the roll cage model. Front impact is one of the major things to be considered in designing a racing car. Most of the competitions have made it mandatory to attach an impact attenuator device at the front portion to withstand maximum amount of force during impact, so that a larger amount of force is absorbed by it. During a full speed impact, the force can be as high as 20,000 N. The necessary input values were given for the analysis. The boundary conditions like fixing the rear end of the vehicle and the load was applied on the front side. The result of the front impact is shown in the Fig. 3(a). The front portion gets crushed and even linkage is broken near the steering portion. The crushing of the front portion is depicted by red colour.

The side impact must also be given considerable attention. It can occur either near driver's cockpit or at the front and rear part of the car. Side impact model with two different materials namely, steel and aluminium was carried out. The side impact at the driver cockpit for model with steel as material is shown in Fig. 3(b). The side impact at the front and rear suspension part of the vehicle for the model with steel material is shown in Fig. 3(b) and Fig. 4(a).

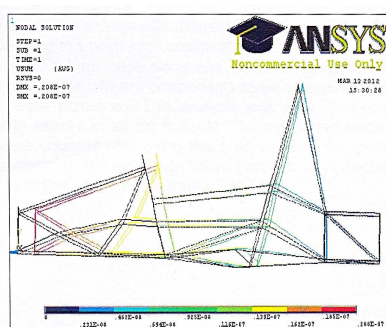
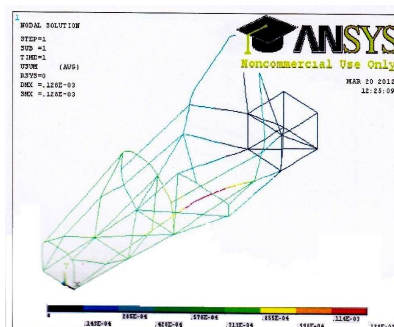
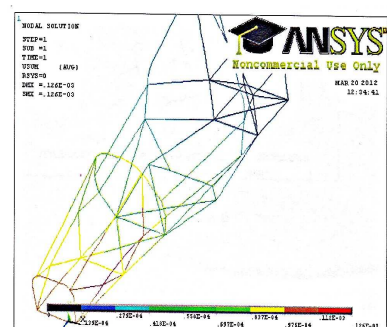


Fig.3 (a) Front Impact;



(b) Side Impact at Driver Cockpit;



(c) Side Impact Front part

The side impact at rear suspension was again tested by changing the material to aluminium. This is shown in Fig. 4(b). It is seen that the impact is very small in case of aluminium compared to that of steel. The analysis results are shown in Table 1.

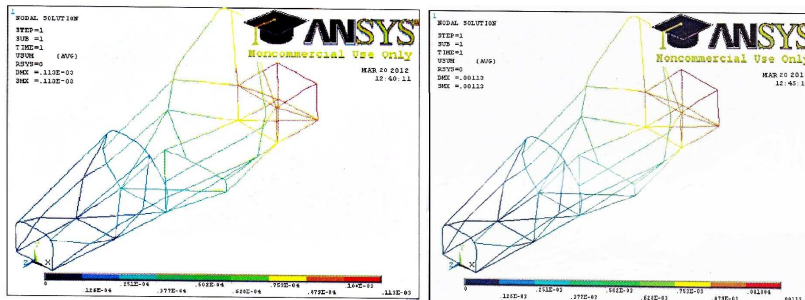


Table 1 Analysis Results

Area of Analysis	Maximum Value
Front Impact Test	0.208E ⁻⁰⁷
Side Impact at Driver Cockpit	0.128E ⁻⁰³
Side Impact at Front Part	0.126E ⁻⁰³
Side Impact at Rear Part	0.113E ⁻⁰³
Side Impact at Rear Suspension	0.00130

Fig. 4 Side Impact (a) Rear Side; (b) Rear Suspension Part

Analysis of Roll Cage Against Longitudinal Torsion

Longitudinal torsion analysis was carried out on the roll cage. A tube of 25 mm diameter and 2 mm thickness was used for the model. Element type used was pipe 16. Material properties are given as input as aluminium whose modulus of elasticity is 70000 Mpa and poisson ratio is 0.33. The constraints are given to the roll cage at the rear end with arresting all degree of freedom. Suspension mounting point is to be displaced in vertical direction. The load was applied at vertical downward of 60 N at each suspension point, one side is vertically downward and other side is vertically upward as shown in Fig. 5(a). Solution is carried out and the deformed shape of roll cage is analyzed. The results show the displacement of frame nodal solution, element solution and bending moment diagram.

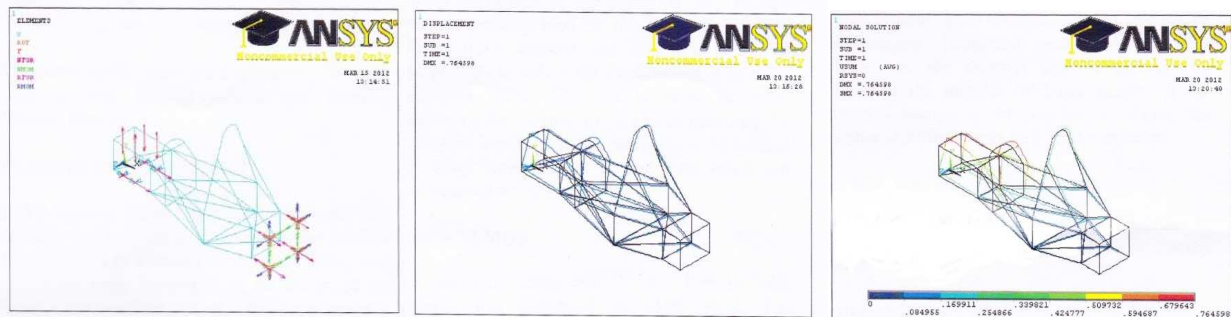


Fig. 5 (a) Boundary Conditions (b) Displacement of Truss; (c) Nodal Solution Diagram;

Table 2 Element and Nodal Values

Minimum Value		
Element No.	519	298
Min. Value	-6566.5	-6566.5
Maximum Value		
Element No.	530	271
Max. Value	4972.5	5041.3
Max. Absolute Value- Nodes		
Node No.	Values	
1534	0.17865	
283	0.44691	
1601	0.76033	
1607	0.76460	

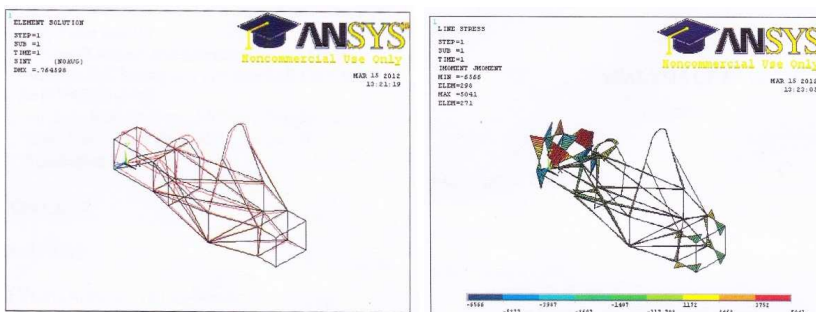


Fig. 6 (a) Elemental Solution: (b) Bending Moment Diagram

Displacement of truss is shown in Fig. 5(b), where a displacement of 0.76460 mm is obtained. This can be clearly seen in the Fig. 5(c) where the nodal solution is shown. The maximum value is obtained in node 1607. The elemental solution as shown in Fig. 6 (a) shows the minimum and maximum values of the shift due to the torsional loading. This is detailed in Table 2. The bending moment is maximum at element number 271 with a value of 5041 N-m and minimum at element number 298 with a value of -6566 N-m.

CONCLUSION

This paper has dealt with a variety of issues related to roll cage design. This study was also conducted to evaluate the driver's risk of injury during a frontal and side impact of a racing car. The impact analysis showed that the car can withstand heavy loads. Both front and side impact analysis were carried out and the maximum and minimum values were tabulated. It was seen that the impact was less if aluminium was used as the material compared to that of steel. Analysis of roll cage against longitudinal torsion was performed. Results showed that the stress on a member can be reduced by increasing the member thickness locally. Increasing the number of stiffener is another option. This would help the designers to develop more competitive roll cage designs.

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